OCTOBER 2026 | VOL. 58 NO. 10 | ISSN-0013-516X

CONTENTS

58TH YEAR OF PUBLICATION
electronics FOR YOU
SINCE JANUARY 1969

TECHNOLOGY

34 DEFENCE: Directed Energy Weapons: Defining Future Warfare





42 INTERVIEW: Components, Constraints, And Market Demands: Navigating The RF Ecosystem — Jin Bains, CEO, Mini-Circuits

48 CAREERS: Edge AI: What Hardware Designers Must Know

SEMICON

58 INTERVIEW: "Government Incentives Can Provide A Runway, But Indian OSATs Must Compete Globally On Their Own Merits" — Venkata Simhadri, MD And CEO, ASIP Technologies

62 INDIA: The Race For Chips Might Become A Race For Water



BUSINESS

81 PURCHASE: Buyers' Guide: Leading Suppliers Of SMT Pick-And-Place Machines

86 MY STORY: Face Setbacks, Learn From Them, And Return Stronger — Malini Narayanamoorthi, President, Renesas Electronics India

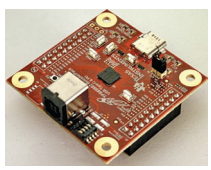


DESIGN

91 CIRCUIT: 5 Interesting Reference Designs For USB-C Power And Charging

93 COMPONENTS: LLMs Are Entering The RF Design Lab

95 SOURCING: Embedded RF Antenna Suppliers: A Complete Guide For Indian Design Teams



DO-IT-YOURSELF

98 PROJECT: Mesh-Controlled Swarm Robots For Thermal Surveys And Analysis

99 CIRCUIT IDEA: On-Delay And Off-Delay Timers Using CD4050 And CD4049

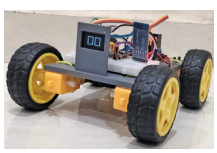
103 PROJECT: Car/Auto Reverse Parking Security System With Audio Video Alert

104 CIRCUIT IDEA: Electronic Pulse Waveform Monitor

105 PROJECT: Pocket-Sized Thermal Camera Using AMG8833

109 PROJECT: Bluetooth-Controlled Robot Using The STM32 Blue Pill

113 PROJECT: Computer-Vision-Based Robotic Arm



REGULARS

10 **Research & Innovation Updates**
20 **Truly Innovative Electronics**
24 **New Products**
46 **Electroons 2.0**
50 **SemiCon:** Pulse
68 **Dashboard:** Key Industry Trends That Can Shape The Future Of Your Business
70 **Innovators & Startups**
78 **Industry News:** Industry Updates

(Cover photos courtesy: https://en.wikipedia.org/wiki)

EDITOR : Rahul Chopra
EDITORIAL CORRESPONDENCE : Phone: +91-9811155335 E-mail: editop@efy.in
SUBSCRIPTIONS & MISSING ISSUES : Phone: +91-9811155335 E-mail: support@efy.in
BACK ISSUES, BOOKS, KITS, COMPONENTS, etc : Phone: +91-9811155335 E-mail: support@efy.in
NEWSSTAND DISTRIBUTION : Phone: +91-9811155335 E-mail: efycirc@efy.in
ADVERTISEMENTS
NEW DELHI (HEAD OFFICE) : Phone: +91-9811155335 E-mail: efyenq@efy.in
MUMBAI : E-mail: rmowest@efy.in
BENGALURU : E-mail: rmosouth@efy.in

CHINA : Worldwide Focus Media E-mail: china@efy.in
GERMANY : PMS Plantenberg Media Service GmbH E-mail: germany@efy.in
JAPAN : Tandem Inc. E-mail: japan@efy.in
TAIWAN : J.K. Media E-mail: taiwan@efy.in
UNITED KINGDOM : ASA Media E-mail: uk@efy.in
UNITED STATES : E & Tech Media E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at Thomson Press India Private Limited, 18/35 Delhi Mathura Road, Faridabad 121007, Haryana, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2026. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

EFY = Electronics For You

PRINT MAGAZINE	1 YEAR (12 copies each)	3 YEARS (36 copies each)	5 YEARS (60 copies each)
WITHIN INDIA (IN RUPEES)			
EFY (Rs 100/ copy)	960	2700	4200
SAARC COUNTRIES (IN US$ BY AIR)			
EFY	57	155	NA
OTHER COUNTRIES (IN US$ BY AIR)			
EFY	114	310	NA

Courier delivery within India @ Rs 50 per copy

e-magazine subscriptions within India are available at half the rates mentioned here.

Overseas rates for each e-magazine in US$:
1 year: $12; 3 years: $33; 5 years: $50 only

Please send payments only in favour of **EFY Enterprises Pvt Ltd** through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: **http://subscribe.efyindia.com** or scan the QR code.